

SHETH LUJ AND SIR MV COLLEGE
Subject: Data Analysis with SAS / SPSS /R

M2 Practical No: 3

Aim: Creating cross-tabulations and two-way tables using `table()` (R).

Code:

1. Clear Environment

```
rm(list = ls())
```

2. Read Dataset

```
cancer_data <- read.csv("Cancer.dataset.csv")
```

3. View Dataset Structure and Data

```
print("--- 3. Data Structure (str) ---")  
str(cancer_data)  
print("\n--- 3. Data Head ---")  
head(cancer_data)
```

4. Data Preparation (Select Paired Variables and Remove Missing Values)

```
data_paired <- cancer_data[c("radius_mean", "radius_worst")]  
  
data_paired <- na.omit(data_paired)
```

5. Paired t-Test

```
paired_ttest <- t.test(  
  data_paired$radius_mean,      # Group 1 (equivalent to Time_Before)  
  data_paired$radius_worst,     # Group 2 (equivalent to Time_After)  
  paired = TRUE,  
  alternative = "less"          # We hypothesize that radius_mean < radius_worst  
)
```

6. Display Result

```
print("\n--- 6. Paired t-Test Results (Mean vs. Worst Radius) ---")  
print(paired_ttest)
```

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Output:

The screenshot displays the RStudio environment with the following components:

- Source Editor:** Contains R code for loading and inspecting the 'cancer' dataset. The code includes:

```
R> R452 - ./ ->  
> read.list = ls()  
> cancer_data <- read.csv("cancer_dataset.csv")  
> print("---- 3. Data Structure (str) ----")  
[1] "---- 3. Data Structure (str) ----"  
> str(cancer_data)  
> data.frame() 569 obs. of 33 variables:  
$ id : int 842302 842517 84300903 84348301 84358402 843786 844359 84458202 844981 84501001 ...  
$ diagnosis : chr "M" "M" "M" "M" ...  
$ radius_mean : num 18.20 19.7 11.4 20.3 ...  
$ texture_mean : num 10.4 17.8 21.2 20.4 14.3 ...  
$ perimeter_mean : num 122.8 132.9 130 77.6 135.1 ...  
$ area_mean : num 1001 1326 1203 386 1297 ...  
$ smoothness_mean : num 0.1184 0.0847 0.1096 0.1425 0.1003 ...  
$ compactness_mean : num 0.2776 0.0786 0.1599 0.2839 0.1328 ...  
$ concavity_mean : num 0.3001 0.0869 0.1974 0.2414 0.198 ...  
$ concave_points_mean : num 0.1471 0.0702 0.1279 0.1052 0.1043 ...  
$ symmetry_mean : num 0.242 0.181 0.207 0.26 0.181 ...  
$ fractal_dimension_mean : num 0.0787 0.0567 0.06 0.0974 0.0588 ...  
$ radius_se : num 1.095 0.543 0.746 0.496 0.757 ...  
$ texture_se : num 0.905 0.734 0.787 1.156 0.781 ...  
$ perimeter_se : num 8.59 3.4 4.58 3.44 5.44 ...  
$ area_se : num 153.4 74.1 94 27.2 94.4 ...  
$ smoothness_se : num 0.0064 0.00522 0.00615 0.00911 0.01149 ...  
$ compactness_se : num 0.049 0.0131 0.0401 0.0746 0.0246 ...  
$ concavity_se : num 0.0537 0.0186 0.0383 0.0566 0.0569 ...  
$ concave_points_se : num 0.0159 0.0134 0.0206 0.0187 0.0188 ...  
$ symmetry_se : num 0.03 0.0139 0.0225 0.0596 0.0176 ...  
$ fractal_dimension_se : num 0.00619 0.00353 0.00457 0.00921 0.00511 ...  
$ radius_worst : num 25.4 25 23.6 14.9 22.5 ...  
$ texture_worst : num 17.3 23.4 25.5 26.5 16.7 ...  
$ perimeter_worst : num 184.6 158.8 152.5 98.9 152.2 ...  
$ area_worst : num 2019 1956 1709 568 1575 ...  
$ smoothness_worst : num 0.162 0.124 0.144 0.21 0.137 ...  
$ compactness_worst : num 0.666 0.187 0.424 0.866 0.205 ...  
$ concavity_worst : num 0.712 0.242 0.45 0.687 0.4 ...  
$ concave_points_worst : num 0.265 0.186 0.243 0.258 0.163 ...  
$ symmetry_worst : num 0.46 0.275 0.361 0.664 0.236 ...  
$ fractal_dimension_worst : num 0.1189 0.089 0.0876 0.173 0.0768 ...  
$ x : logi NA NA NA NA NA ...  
> print("\n---- 3. Data Head ----")  
[1] "---- 3. Data Head ----"
```
- Environment Pane:** Shows the loaded objects: 'cancer_data' (569 obs. of 33 variables), 'data_paired' (569 obs. of 2 variables), and 'paired_ttest' (List of 10).
- Console:** Displays the output of the R commands, including the structure of the data frame and the first few rows of the dataset.
- Files Pane:** Shows the project files, including 'cancer_data.csv' and 'data_paired.csv'.

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The screenshot shows the RStudio interface with the following components:

- Source Editor:** Contains R code for data manipulation and a paired t-test.
- Console:** Displays the output of the R code, including the results of the paired t-test.
- Environment:** Shows the objects in the global environment: `cancer_data` (569 obs. of 33 variables), `data_paired` (569 obs. of 2 variables), and `paired_ttest` (List of 10).

```
R - R4.5.2 - ~/r
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function Addins
Source
Console Terminal Background Jobs
R - R4.5.2 - ~/r
6 0.007510 0.03345 0.03672 0.01137 0.02165 0.005082 15.47
texture_worst perimeter_worst area_worst smoothness_worst compactness_worst concavity_worst concave.points_worst
1 17.33 184.60 2019.0 0.1622 0.6656 0.7119 0.2654
2 23.41 158.80 1956.0 0.1238 0.1866 0.2416 0.1860
3 25.53 152.50 1709.0 0.1444 0.4245 0.4504 0.2430
4 26.50 98.87 567.7 0.2098 0.8663 0.6869 0.2575
5 16.67 152.20 1575.0 0.1374 0.2050 0.4000 0.1625
6 23.75 103.40 741.6 0.1791 0.5249 0.5355 0.1741
symmetry_worst fractal_dimension_worst X
1 0.4601 0.11890 NA
2 0.2750 0.08902 NA
3 0.3613 0.08758 NA
4 0.6638 0.17300 NA
5 0.2364 0.07678 NA
6 0.3985 0.12440 NA
> data_paired <- cancer_data[c("radius_mean", "radius_worst")]
> data_paired <- na.omit(data_paired)
> paired_ttest <- t.test(
+ data_paired$radius_mean, # Group 1 (equivalent to Time_Before)
+ data_paired$radius_worst, # Group 2 (equivalent to Time_After)
+ paired = TRUE,
+ alternative = "less" # We hypothesize that radius_mean < radius_worst
+)
> print("\n--- 6. Paired t-Test Results (Mean vs. Worst Radius) ---")
[1] "\n--- 6. Paired t-Test Results (Mean vs. Worst Radius) ---"
> print(paired_ttest)

Paired t-test

data: data_paired$radius_mean and data_paired$radius_worst
t = -30.801, df = 568, p-value < 2.2e-16
alternative hypothesis: true mean difference is less than 0
95 percent confidence interval:
 -Inf -2.027327
sample estimates:
mean difference
 -2.141898
>
> |
```